So You Think You Can DAS

A guide to annotating whale calls with `annotate_das`

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last updated: November 10, 2026

1 Introduction

This document serves as an introduction and tutorial on the `annotate_das` Graphical User Interface (GUI). `annotate_das` is used to annotate whale calls in Distributed Acoustic Sensing (DAS) data across several related domains:

- **T–X plot**: time versus cable distance.
- **F–X plot**: frequency versus cable distance for a selected short time window.
- **Spectrogram**: frequency versus time at one selected cable distance.
- **F–X thumbnail series**: scrollable collection of short-time F–X windows.

Each of these views is described in more detail in section 3.1.
The annotation workflow records:

- apex time and cable distance;
- endpoint time;
- roughly estimated source distance from cable (used to map between domains, not sufficiently precise for localization);
- approximate sound speed (used to map between domains, not sufficiently precise for localization or analysis);
- selected minimum and maximum cable distance;
- selected frequency band;
- label (species, call type);
- source file, UTC timing information, annotator.

All absolute timestamps are stored and displayed in UTC. The source distance and sound speed are used to generate the hyperbola shape, and are selected based on visual alignment of a hyperbola to the observed data. These are not rigorous or reliable estimates of the true offset distances or sound speeds, and should only be used to generate masks or to map labels between views.

2 Getting Set Up with `annotate_das`

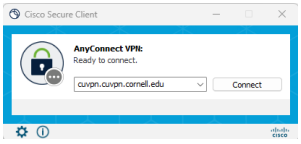
2.1 VPN and Remote Connection

Connect to eduroam with `netID@cornell.edu` and your password.

2.1.1 Connect to the VPN

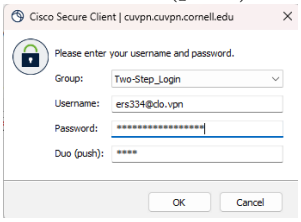
To connect to the VPN:

1. Cisco AnyConnect should already be installed. Open it, and ensure that `cuvpn.cuvpn.cornell.edu` is entered in the text box (it should be by default). Click "Connect".



2. Enter your Cornell ID as the username followed by @clo.vpn (e.g. `abs123@clo.vpn`) and your Cornell password int he "Password" box.

3. In the "Duo (push):" box, type `push`.



4. AnyConnect should now send an approval request to Duo Mobile on your phone. Accept this and you will be connected to the VPN. AnyConnect will close, but it is still connected in the background.

2.1.2 Connect to the desktop

To connect to the remote computer:

1. Open Remote Desktop Connection.

2. In the "Computer" box, type the address of the computer you wish to connect to:

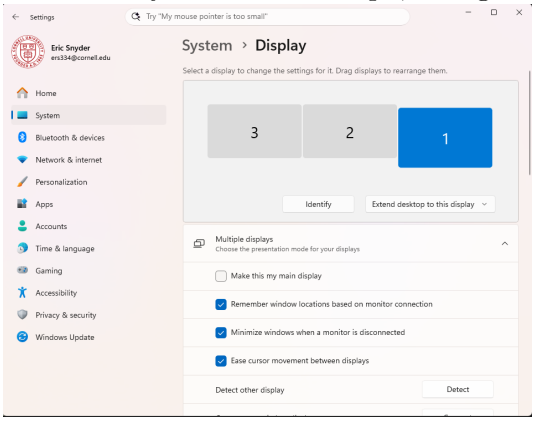
   - `ag-clo-929w7b3.ad.cornell.edu` (Michael and Emily)
   - `ag-clo-92cd3h4.ad.cornell.edu` (Isabella)

3. Click "Connect", enter your password, and click "OK".

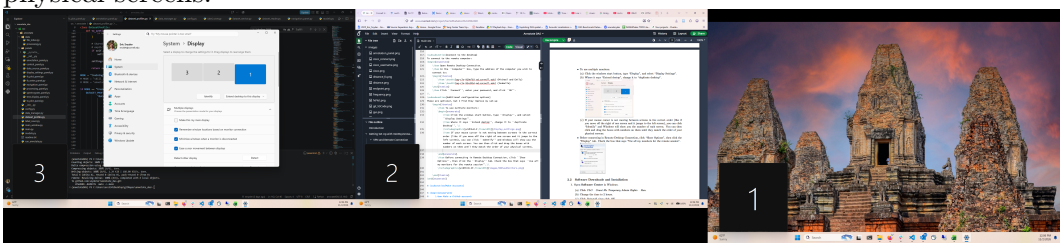
2.1.3 Additional configuration options

These are optional, but I find they improve my set-up

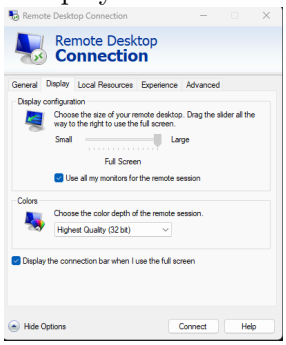
- To use multiple monitors:
  (a) Click the windows start button, type "Display", and select "Display Settings".
  (b) Where it says "Extend destop", change it to "duplicate desktop".



  (c) If your mouse cursor is not moving between screens in the correct order (like if you move off the right of one screen and it jumps to the left screen), you can click "Identify" and Windows will show you the number of each screen. You can then click and drag the boxes with numbers on them until they match the order of your physical screens.



- Before connecting in Remote Desktop Connection, click "Show Options", then click the "Display" tab. Check the box that says "Use all my monitors for the remote session".



2.2 Software Downloads and Installation

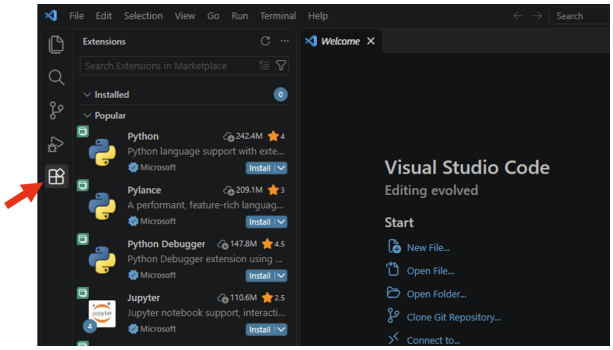
1. Open **Software Center** in Windows.
   (a) Click *CLO – Grant Me Temporary Admin Rights – Run.*
   (b) Change the time to 2 hours.
   (c) Click *Reinstall*, then click *OK.*
   (Note: You must do this every time you install new software. You should now have administrator rights and be able to install software for two hours.)

2. Install Visual Studio Code (VSCode).

   (a) Download VSCode from: `https://code.visualstudio.com/download`

   (b) Log in using your GitHub information.

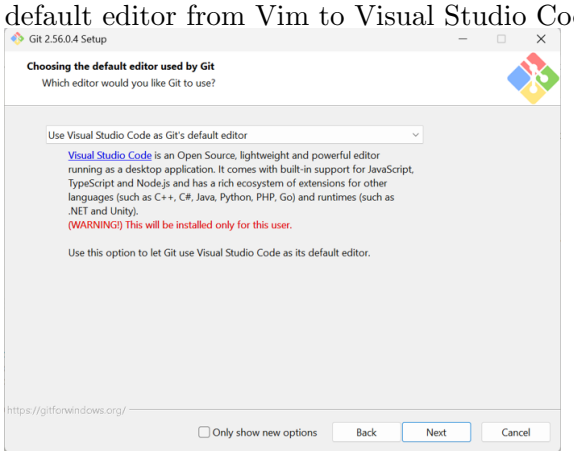
   (c) Click *Extensions* in the left-hand sidebar and install the *Python* extension.



3. Install Git.

   (a) Download the Git installer from the official Git SCM download page . Select the *64-bit Git for Windows Setup* standalone installer.

   (b) Click *Next* and retain all default settings, **except for the default editor**. Change the default editor from Vim to Visual Studio Code.



4. Configure your Git identity.

   (a) Close and reopen VSCode. Press `Ctrl + ‘` to open the integrated terminal.

   (b) Verify that Git is working:

```
git --version
```

   This should print output similar to:

```
git version 2.56.0.windows.2
```

   (c) Set your Git user name:

```
git config --global user.name "Your Name"
```

(d) Set your Git email address:

```
git config --global user.email "your.email@example.com"
```

5. Install Miniconda.

   (a) Download Miniconda. If the embedded download link does not work, visit: `https://www.anaconda.com/download`. Scroll to the bottom and click "miniconda". Under the *Miniconda* tab, select *Windows 64-Bit (x86_64) Graphical Installer*.

   (b) Install Miniconda using all default settings. **Make note of the file path where Miniconda is installed.**

   (c) To Activate Miniconda, open PowerShell and enter the following command, replacing the example path with your own Miniconda path:

```
& "C:\Users\ers334\AppData\Local\miniconda3\Scripts\conda.exe" init powershell
```

   (d) Close PowerShell and VSCode.

2.3 Install and Set Up the `annotate_das` Software

1. Get the `annotate_das` software.

   (a) Reopen VSCode and open a terminal (`Ctrl + ‘`). Navigate to the Desktop in the VSCode terminal:

```
cd Desktop
```

   (b) Create a new folder named `DAS`:

```
mkdir DAS
```

   (c) Move into the new folder:

```
cd DAS
```

   (d) Clone the `annotate_das` Git repository:

```
git clone https://github.com/snyderer/annotate_das.git
```

   (e) Click File → Open Folder or press `ctrl + k + o`. Click on `annotate_das` then click "Select folder".

2. Set up the Conda environment.

   (a) Create and activate the environment:

```
conda create -n annotate_das python=3.14 pip -y
conda activate annotate_das
```

   (b) Install the required Python packages:

```
pip install -e .
```

   If an error indicates that `pip` is unrecognized, install `pip` into the Conda environment:

```
conda install pip
```

Then run the package-installation command again:

```
pip install -e .
```

3. Try running the graphical user interface (GUI):

```
python .\src\annotate\main.py
```

4. If the GUI does not run, check whether the error messages identify missing packages and install them as needed. Likely missing packages include:

```
pip install das4whales
pip install git+https://github.com/qgoestch/simpleDAS
```

3 What am I looking at? A brief description of DAS, data, and plots.

Acoustic data is (usually) a measurement of changes in pressure over time. When somebody uses a microphone or hydrophone (underwater microphone) to record an "acoustic waveform", they are usually using a device which converts small changes in pressure due to a propagating acoustic wave into a voltage. **Distributed Acoustic Sensing** (DAS) is a little more complicated. First, because we have an **entire cable** over which we are measuring an acoustic signal; this means we are measure the acoustic waveform in time AND space, so typical methods for displaying and analyzing acoustic data don't always work. Second, DAS doesn't directly measure changes in pressure due to an acoustic wave; instead, it measures **strain** wave; instead, it measures **strain**[1]: how much the cable is stretching or compressing ...: how much the cable is stretching or compressing due to vibrations induced by an acoustic wave. The transfer of acoustic energy into strain is not always a straightforward conversion, and creates several unique characteristics in the data (to be discussed later). Before we get to that, however, we need to talk about plots.

3.1 Plots

We display data in three formats: **T-X plots**, **F-X plots**, and **Spectrograms**. Each of these displays different features of the data, and each one is useful for exploring, identifying, verifying, and annotating data.

[1] To make things more complicated, DAS actually only measures along-cable strain, *i.e.*, the component of the strain that is along the same axis as the cable[a].

[a] To make things even more complicated, it's *technically* not even directly measuring strain — DAS measures phase differences in Rayleigh back-scattered light that are induced by along-cable strain, but this is more than you need to know right now.

3.1.1 T–X plots

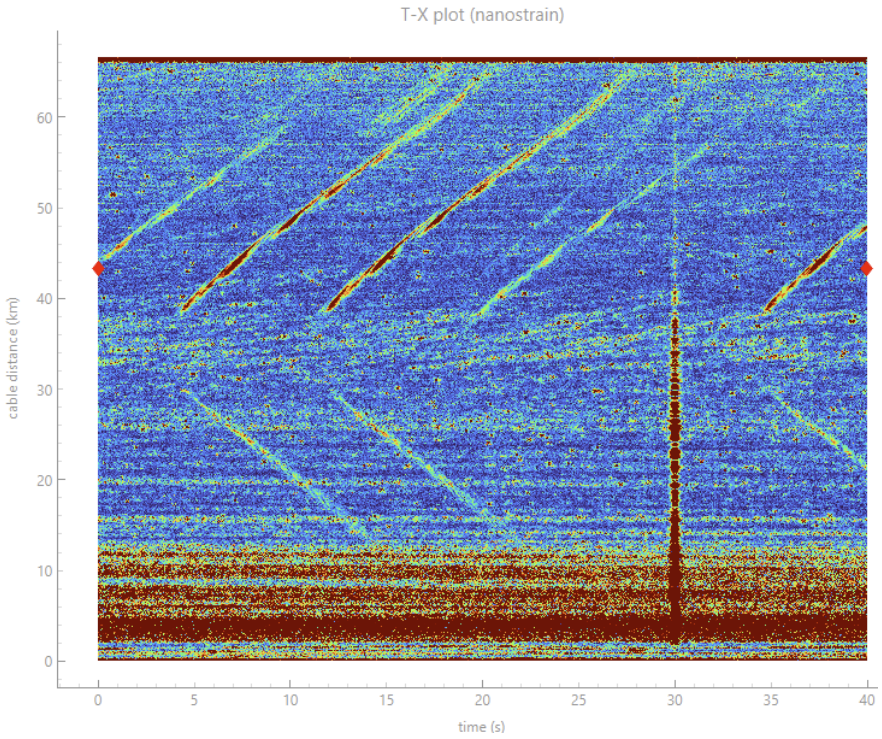


Figure 1: An example of a T-X plot with axes of time (T) and distance (X). The color scale represents the strain of the cable, which corresponds to the intensity of the acoustic signal, where the color goes from blue to green to red as the strain gets more intense.

The most intuitive way to display DAS data is in a "T–X plot", with time on the x-axis and cable distance on the y-axis, and the intensity of the "strain" as the color scale. X refers to the distance axis and T refers to the time axis [2] (Figure 1).

T-X plots allow you to quickly examine a window of data for interesting signals. Figure 1 shows 40 seconds of data across a ≈ 67 km section of cable – an enormous chunk of data being displayed all in one image! In this example, you can immediately see several features:

- There is a lot of energy from 0 to 13 km that seems to present throughout the 40 seconds. This is caused by shallow-water ocean waves or human-generated sound and is not particularly of interest to us. Since those signals dominate the smaller distances, this portion of the cable is practically useless to us. (**Magenta box** on Figure 2)
- The vertical line at 30 seconds is an artifact of the data loading and can be ignored[3]. You can tell it's not an actual sound propagating through the water because it arrives at every channel simultaneously. This would require the sound to be propagating through the water at an infinite speed. (**Green box** on Figure 2)
- The sloped lines *are* in fact signals of interest. These are fin whale song! You can see that the slopes of those lines are pretty consistent. Since they are propagating at the speed of sound

[2] Yes, this can get confusing because T is on the x-axis and X is on the y-axis.

[3] DAS datasets are far too large to be stored in one file. In this example, each file is 30 seconds long. The energy spike at 30 seconds occurs every time a new file is loaded into the GUI.

in water (around 1500 meters per second), you are actually seeing the sound travel down the cable. The slope of line is determined by the sound speed. **Looking for slopes like this is a good way to identify actual acoustic signals.** (**White boxes** on Figure 2)

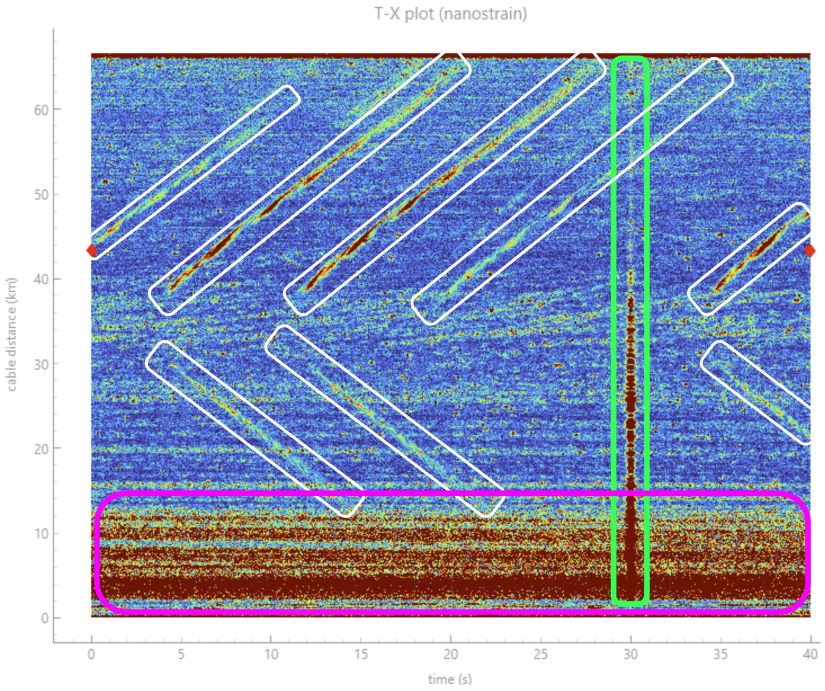


Figure 2: Some observable energy in typical T–X data.
The **magenta box** shows shallow-water ocean-wave noise.
The **green box** shows noise from data anomalies.
The **white boxes** show fin whale song.

Figure 3 highlights a single fin whale call propagating along the cable. A fin whale sat at around 35 km along the cable, and some unknown distance away from the cable. He made a sound which began hit the 35 km section of the cable around the 11 second mark. The call is then observed in both directions along the cable as it continues it's propagation path through the water. We call the point in time and space where the energy first hits the cable the **apex**. You might notice a significant gap in energy near the apex – there is very little energy between 30 and 38 km. This *should* be the region closest to the whale, where acoustic energy is the highest. In fact, if you placed a hydrophone right at the 37 km position on the cable, it *would* detect higher acoustic energy than at any other place along the cable, but DAS is different. DAS has some physical and technical constraints which significantly diminish its ability to measure acoustic energy near the apex. The size of this gap depends on characteristics of the device used to collect the data, how far the whale is from the cable, and the frequency of the whale call. If you're really curious, I can sit with you to explain all that sometime, but for now just know that you'll need to match up the energy on either side of the apex for many whale calls.

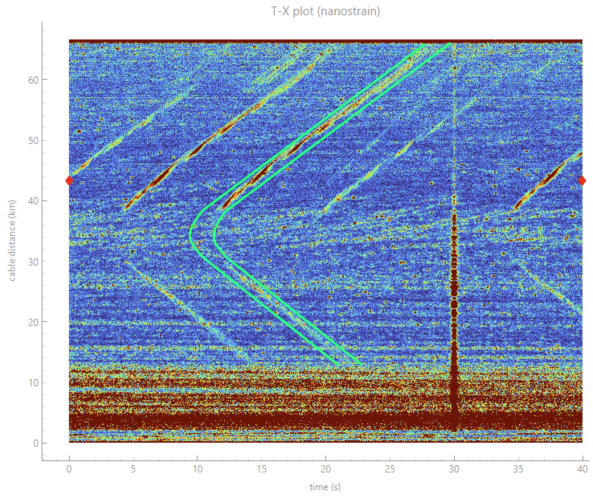


Figure 3: The green curve outlines an individual fin whale call. The whale call starts at a cable distance of about 35 km and travels through the water in both directions along the cable. The gap in acoustic energy near the earliest arrival in time (around 11 seconds) (aka the **apex**) is a well-known feature of DAS data.

I said earlier that the slope of energy is dependent on the speed of sound. For whales that are very close to the cable, the T–X plots will show an almost perfect < shape. Calls that are further away will have a more curved shape (. The further the whale is from the cable, the more curved that shape will be. This is due to the way sound propagates – sound moves more or less in a spherical shape away from the source, so what we are actually seeing is an ever-expanding sphere of acoustic energy moving through the water and hitting the cable. If you work out all the geometry of when sounds will arrive at different sections of the cable, it forms a shape known as a **hyperbola**. Hyperbolae can have different amounts of curvature – sounds originating from a whale that is close to the cable will have almost no curvature, and sounds from whales that were very far away will more curvature.

Therefore, overall, the distance between the whale and the cable has three effects on the way the sound looks in the T–X plot:

1. Sounds that are further away will often be quieter. Sound energy diminishes with distance, so less energy will arrive at the cable.

2. Sounds that are further away will have more curvature in the T–X plot.

3. Sounds that are further away will also have a wider gap near the apex. s

3.1.2

Spectrograms are a common way to display acoustic data. The axes of a spectrogram are time on the x-axis, frequency on the y-axis, and intensity as the color scale. Frequency are typically displayed in **Hertz** (abbreviated Hz), which is the number of oscillations per second. Spectrograms are great for displaying which frequencies are most prominent over time.

The spectrogram in Figure 4 shows the same fin whale song we saw in Figure 2. Fin whale song is a "downsweep", meaning it starts at a higher frequency and goes lower over time (usually from around 25 Hz to around 17 Hz). These are often called 20 Hz calls. They are referred to as "song" because they are made only by males and seem to serve a function in mating, perhaps as a display

of fitness to attract a mate. If we had only observed the data in the T–X plot, we would have been unable to identify the frequencies present in the data, which would have made identifying the call type difficult. Most studies use spectrograms to identify call type and species, so this view is convenient for comparing sounds to existing literature.

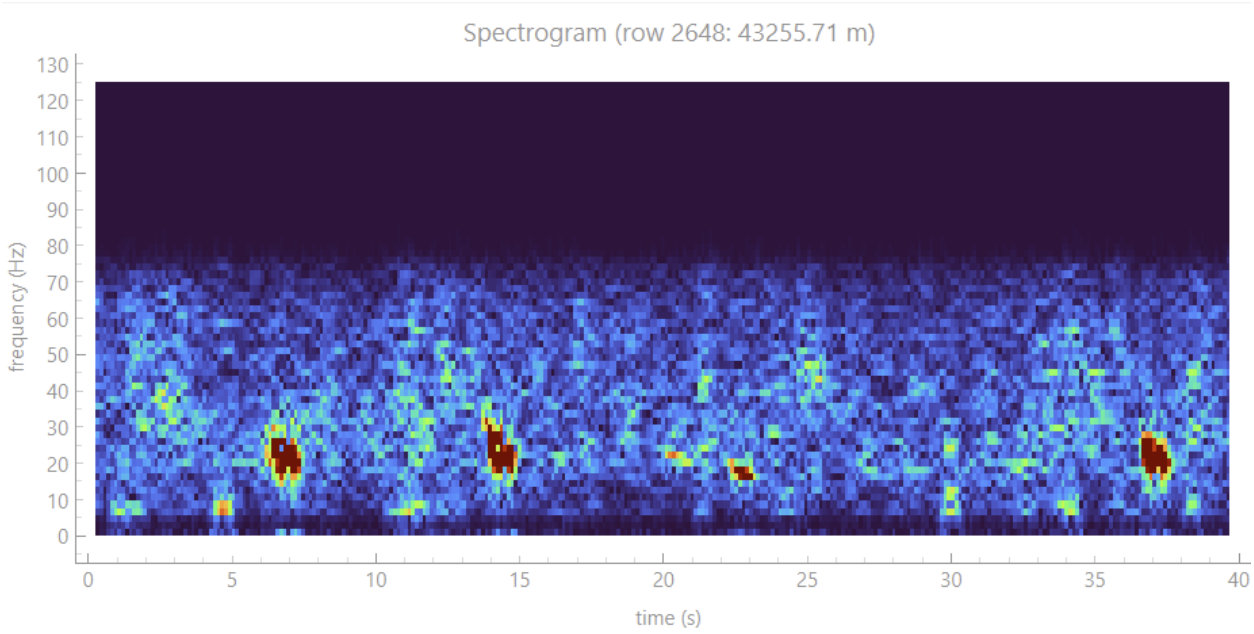


Figure 4: An example of a spectrogram displaying fin whale song. Several calls can be observed in this time-frame, all centered around 20 Hz. The first call occurs at about 7 seconds, the second at 14 seconds, and so on.

Spectrograms have a major limitation when displaying DAS data: One spectrogram can only show data from one channel. Spectrograms are very useful to see how the frequency content is changing over time, but you can't as easily observe the energy moving through space.

There are several parameters that can be selected when generating a spectrogram. You shouldn't have to worry much about that for now, but there are settings you can adjust when you get more comfortable with acoustic and DAS analysis.

3.1.3 F–X plots

F–X plots show the frequency (F) content across distance (X) (Figure 5. Like spectrograms, this is a useful view for identifying whale call energy, but rather than changes over time, you are observing changes across space. As such, this can be a more reliable view for identifying true whale calls rather than spurious noise. For these plots, I generally put frequency on the x-axis and distance on the y-axis to better align the distance axes between F–X and T–X plots 6.

Calculating the frequency content requires a data from a window of time – the longer the window of time, the better your frequency resolution can be, but you risk minimizing the visibility of shorter-duration signals. I have found that a window length of two seconds is a nice sweet-spot for observing our signals, which is the default window length in the `annotate_das`. So, while the T–X plot displayed 40 seconds of data, the F–X plot shows the frequency content from only a 2-second section of the T–X plot.

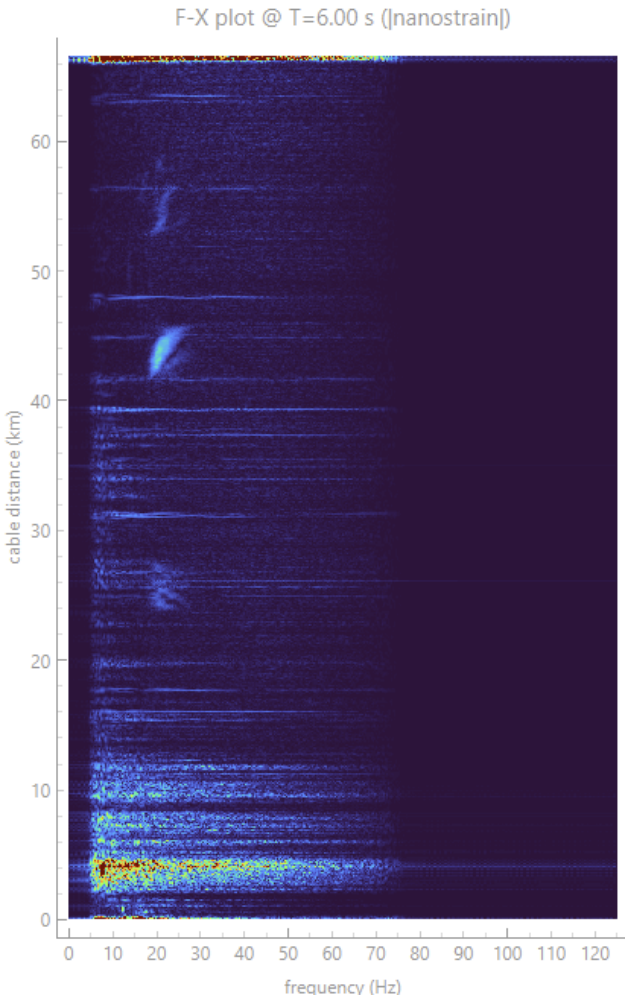


Figure 5: An example of an F–X plot, with frequency (F) on the x-axis, and distance (X) on the y-axis.

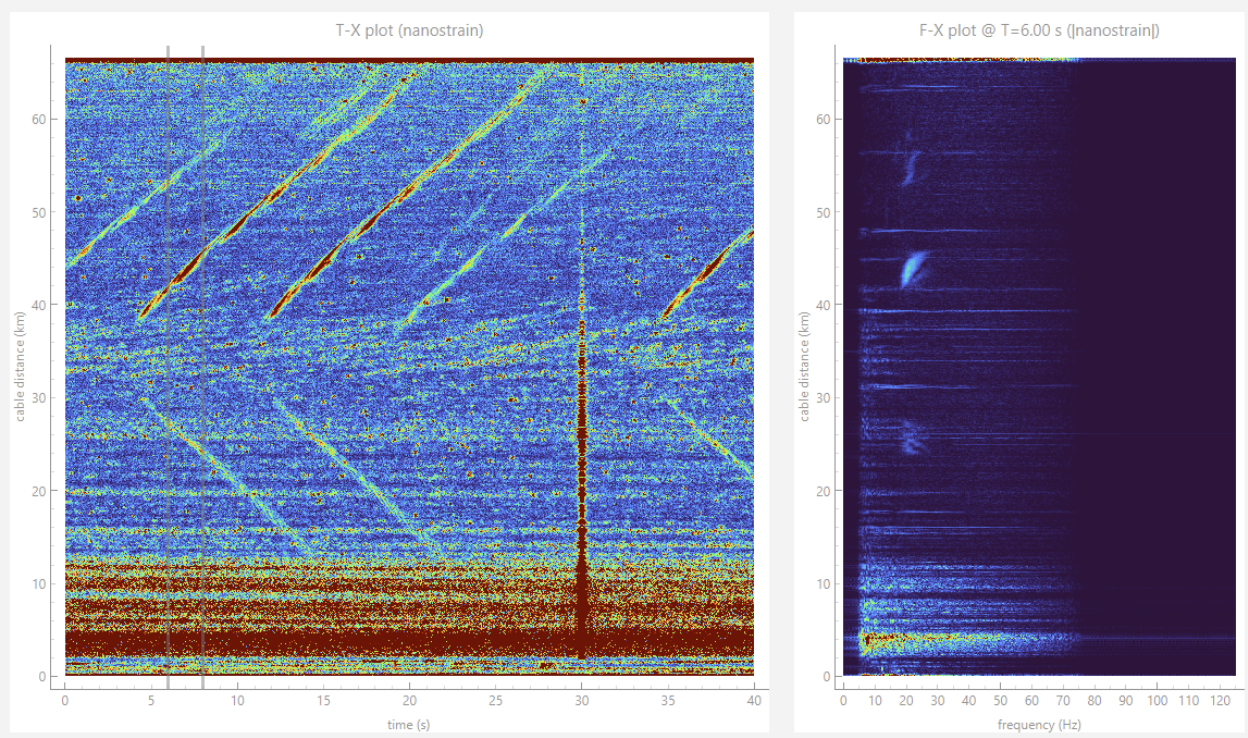


Figure 6: A T–X and F–X plot placed side-by-side, showing the distance axes alignment. The gray vertical lines from 6-8 seconds

Table 1: Interface layout and panel descriptions.

Location	**Contents**
Left panel	**Control panel**: Data source, navigation, annotation information, processing settings, and display settings. The most important sections are navigation and annotation information.
Upper middle-left	**T–X plot**: Displays the T–X data for the window defined in the Navigation time. It also contains indicators for which row is displayed in the spectrogram plot (♦ red diamonds) and which window of time is displayed in the F–X plot (‖ gray vertical lines).
Upper middle-right	**Main F–X plot**: The F–X plot for the window of data indicated by the ‖ gray vertical lines on the T–X plot.
Lower middle-left	**Spectrogram**: The spectrogram for the row indicated by the ♦ red diamonds on the T–X and F–X plots.
Lower middle-right	**Information and status panel**: Shows messages relating to the data and annotation, including "Cursor Mode", dataset information, metadata, and timing information.
Right panel	**F–X thumbnail series**: Displays the F–X plots for the entire duration of the T–X plot. You can click on any thumbnail in this panel and it will show a larger version of that plot in the F–X plot panel. The currently displayed window is boxed in blue.

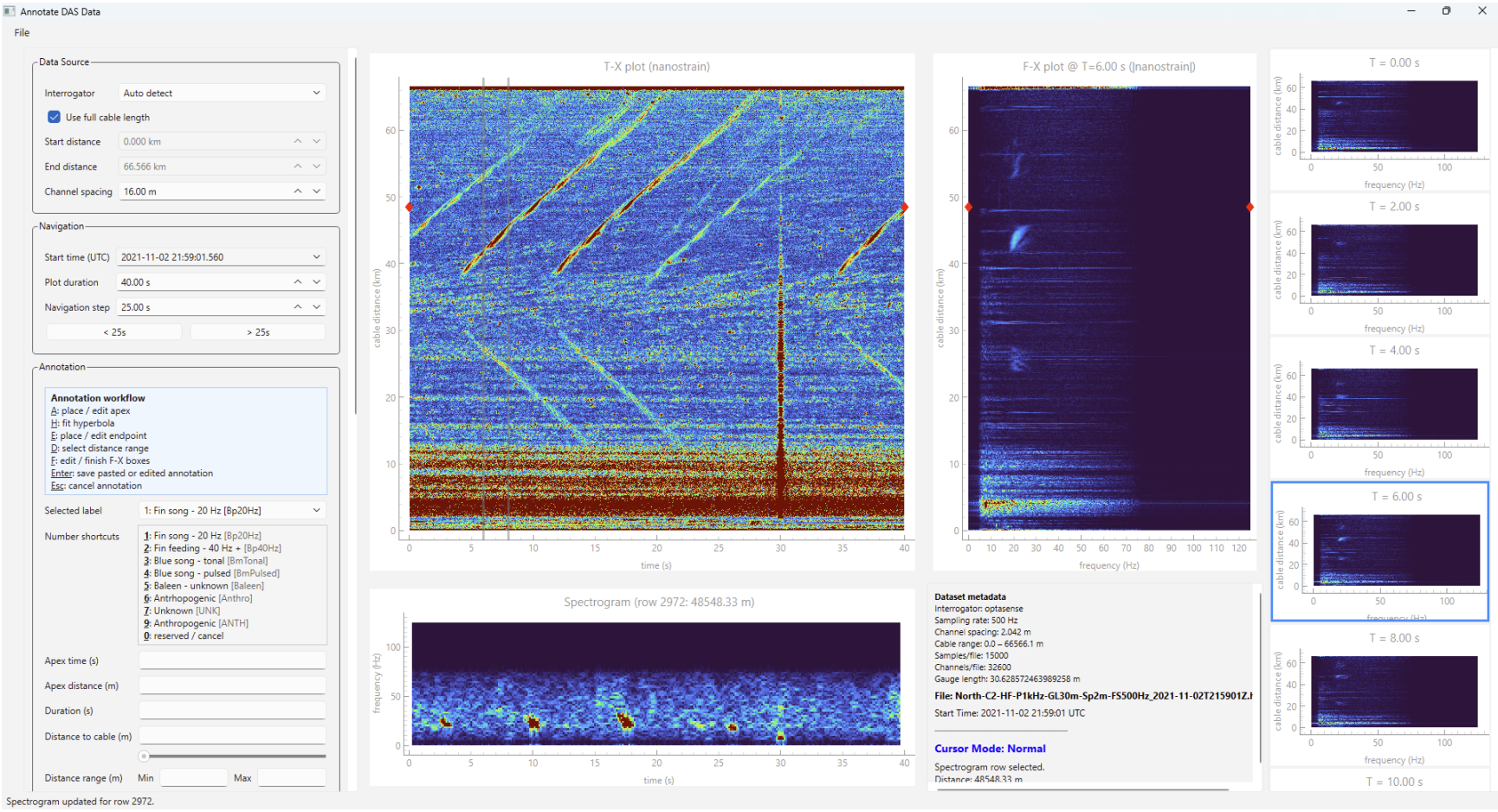


Figure 7: The `annotate_das` interface.

4 Interface Layout

The `annotate_das` GUI is divided into six panels (Table 1; Figure 7). Most of the panels display plots which were described in Section 3.1. A few panels merit further description.

4.1 Control Panel

The control panel contains the data-source controls, navigation settings, annotation information, processing settings, and display settings.

4.2 Control Panel

The **control panel** contains controls for navigating data, adjusting annotations, or manually adjusting settings for the data. The panel is divided into several sections, which are described below.

4.2.1 Data Source

Data Source allows users to select the interrogator type and which channels they want to be loaded and displayed. **This will be set by default when you select a dataset and you do not need to change it.**

4.2.2 Navigation

The **Navigation** section contains controls for navigating through the data. The **Start time (UTC)** box shows the time stamp associated to $T = 0$s on the T–X plot, **Plot duration** sets the full length of time that is displayed, and **Navigation step** shows how many seconds forward/backward you will move when you press the navigation buttons. These are all set automatically, so generally you should leave them at their default values. However, you may occasionally need to adjust these values if you can't adequately annotate a whale call with the displayed parameters. If a whale call seems to continue onto the next window, but navigating by the Navigation step size will cause the beginning of the call to be too early, you should select parameters which will allow the entire whale call to be displayed in one T–X window. See Figure 8 for an example.

4.2.3 Annotation

The Annotation section (Figure 9 displays the **Annotation workflow**, which shows you the series of keystrokes you should use to annotate a whale call. Most of the subsequent boxes will be automatically populated as you go through the annotation steps outlined in Section 5.2. The values can be manually changed in these boxes, but generally only the **Distance to cable (m)** slider will need to be adjusted.

4.2.4 Processing and Display

The Processing and Display sections will be set automatically and you should not have to adjust these. However, if you are exploring data beyond the annotations, these may be useful. They include things like filters (both bandpass and f–k filters), F–X plot window lengths (default 2 seconds), Spectrogram parameters, and plot color scale sensitivity. I wouldn't recommend messing with this unless you know what you're doing.

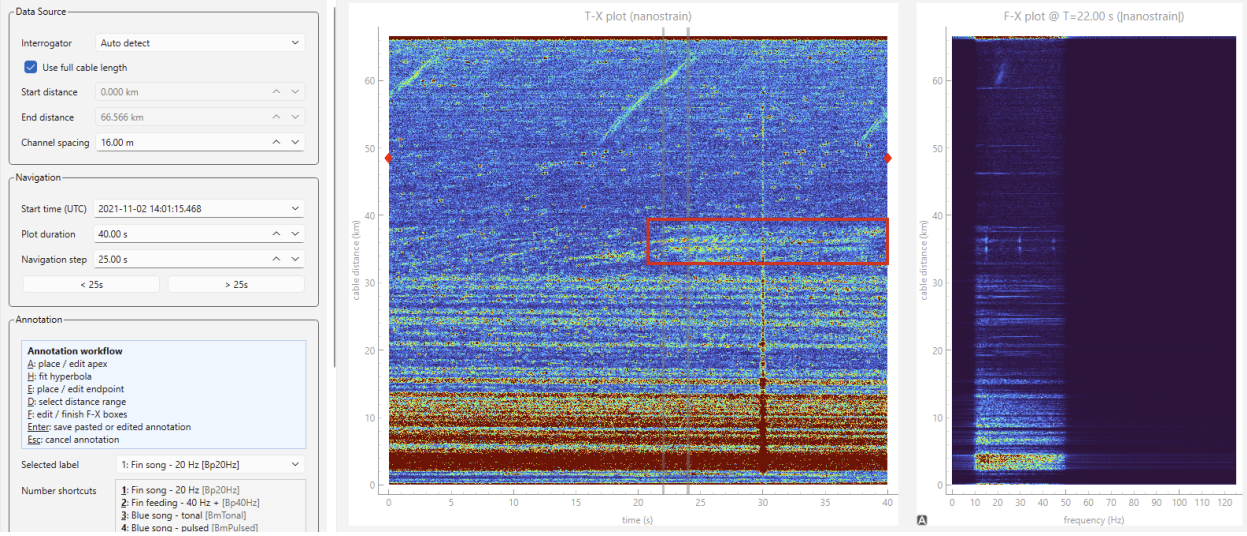


Figure 8: An example of a whale call which will require you to adjust the parameters in the Navigation section. A blue whale pulsed call can be seen at the 34 to 37 km section of the cable (inside the red rectangle). It starts at about 22 seconds and continues until the end of the 40 second window. You wouldn't be able to label the entire call because it is not fully contained on your T–X plot. However, if you were to click "¿ 25s", the beginning of the call would start before the beginning of the displayed window. Instead, you'll have to adjust the values in the navigation boxes to better capture the full call. Start by changing **Navigation step** to 18 or 19 s, then click next. This should navigate to about 3 seconds before the call starts. If the end of the call is *still* past the edge of the window, you will have to adjust the **Plot duration** until the whole call can be seen.

5 Annotating whale calls

To run, open VSCode and start a terminal `Ctrl + ‘`. Click "File -¿ Open Folder", and navigate to `C:\Users\YourUsername\Desktop\DAS\` and select `annotate_das`. Then, in the terminal type:

```
conda activate annotate_das
python .\src\annotate\main.py
```

5.1 Open a Dataset

Use:

File → Select Dataset...

Select the dataset you want to load and annotate. **Loading data can take a long time, so be patient.** The GUI will be unresponsive and show "Not Responding" in the top-left title bar. This is normal and I'm trying to fix it. Just trust that it will load eventually. If you have waited more than 1 or 2 minutes, check VSCode's terminal to see if there's an error message. If there is an error message, send it to Eric in Slack, then try closing the GUI and restarting it by typing `python .\src\annotate\main.py` in the terminal.

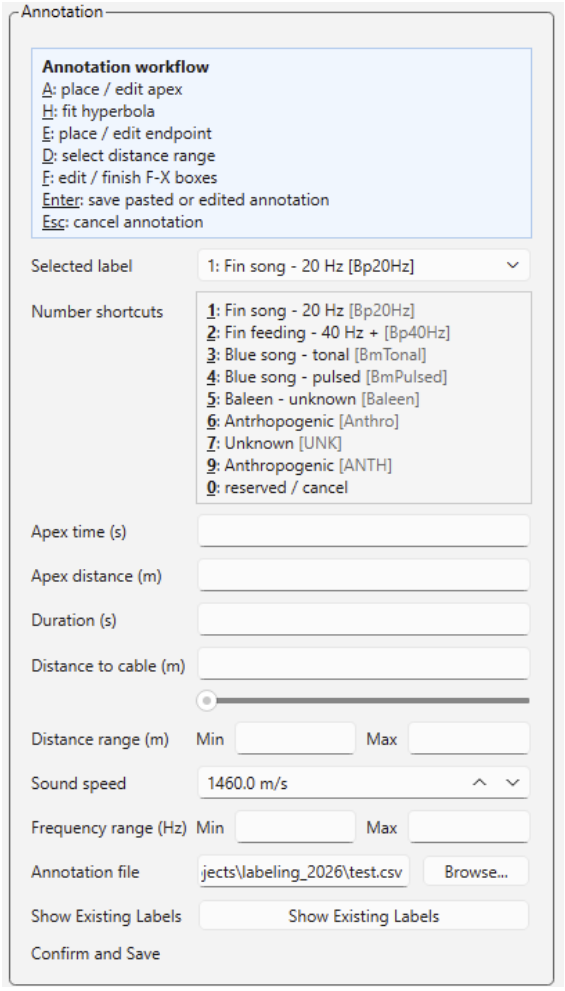


Figure 9: The annotation panel.

5.2 Annotation Workflow

The annotation workflow is initiated by pressing `a`. In general, the steps are:

`a`	Select apex, then confirm apex.
`h`	Fit hyperbola, then confirm source distance to cable.
`e`	Select endpoint, then confirm endpoint.
`d`	Select cable-distance range, then confirm range.
`f`	Create/edit F–X boxes, then finish F–X annotation.
`1--9`	Assign and save numeric label.
`Esc`	Cancel current annotation.

The Annotation panel provides a visible reminder of these shortcuts.

5.2.1 Step 1: Select the Apex

Press:

`a`

The application enters apex-selection mode.
Click the energy apex in the T–X plot. The apex marker is shown in red.
You may:

- click another location to replace the apex;
- drag the red apex marker to refine its position.

The Annotation panel updates with:

- **Apex time**: absolute UTC timestamp;
- **Apex distance**: cable distance in meters.

5.2.2 Step 2: Fit Distance to Cable

Press:

`h`

`h` stands for **Hyperbola**, the shape the calls make in the T–X plots. The **Distance to cable** slider then becomes active. A white dashed hyperbola appears in the T–X plot and is anchored at the selected apex.

Adjust the slider until the white hyperbola follows the observed arrival geometry. The slider represents the distance from the source to the cable, in meters. You can also click and drag the apex around until it best fits the data.

A good fit at this stage is to be as close to the earliest energy you see across all channels, but always make sure the white hyperbola is *before* the whale call. This serves as the **left boundary**.

Scan along the whale call to make sure none of the whale call energy arrives prior to the white hyperbola. If the energy is arriving prior, drag your apex to the left or change your distance slider until it fits.

Try to get the white hyperbola as close to the beginning of the energy as possible without having any whale-call energy to the left of it. We don't want a lot of non-whale call energy inside our segment, but if you have to choose between including a little more non-whale call energy or excluding a little bit of the whale call, it is better to include non-whale call energy.

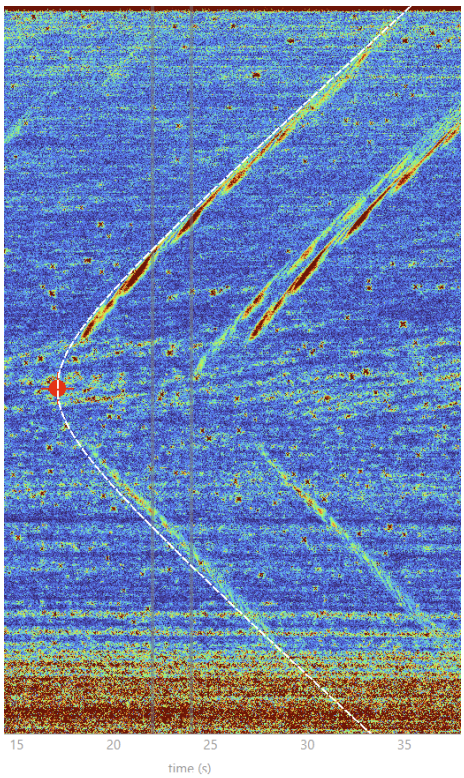
5.2.3 Step 3: Select the Endpoint

Press:

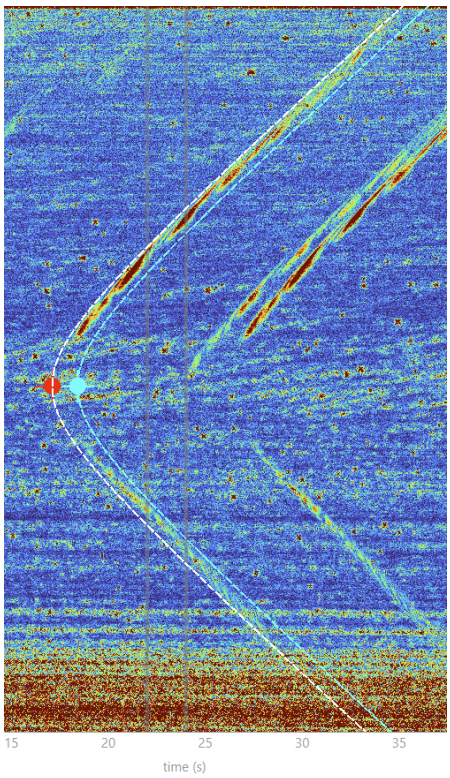
`e`

Click a point to the **right** of the whale call to indicate where the call ends. You must click in approximately the same row as the apex (a.k.a. the same distance along the cable), but later in time. The endpoint is displayed as a cyan marker, and will show a hyperbola representing the end of the whale call across all channels. Make sure the entire whale call fits between the white and cyan hyperbolae.

You may drag the cyan endpoint marker horizontally to refine its position time. It will remain locked to the same cable distance as the apex.



(a) A correctly marked apex and hyperbola. The whale-call energy arrives entirely *after* the white dotted hyperbola, and the apex (red dot) is approximately centered between the two sides of the call.



(b) A correctly placed endpoint. All whale-call energy lies to the left of the cyan hyperbola, so the full call lies between the white and cyan hyperbolae.

Figure 10: Examples of selecting the apex/hyperbola and endpoint during annotation.

6 Step 4: Select Cable-Distance Range

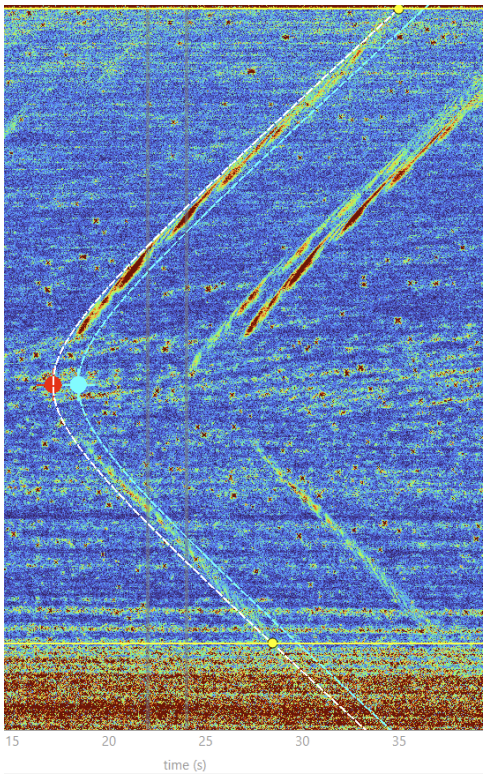
Press:

`d`

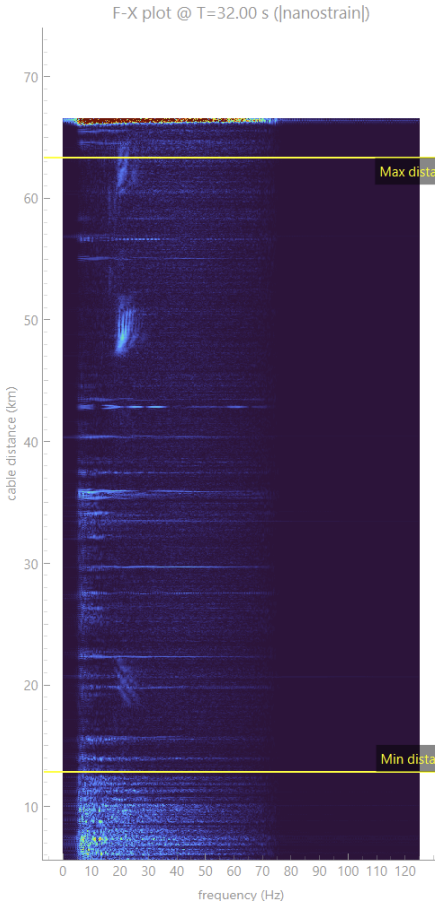
Click the lowest and highest points where you can see whale call energy on the T–X plot. These represent the distances along the cable where the whale call is visible. Each point snaps to the nearest point on the yellow hyperbola.

Once both distance points are placed, two horizontal yellow lines will appear on the T–X and F–X plots. The F–X distance lines are draggable. Dragging one line adjusts that distance boundary and updates the corresponding T–X point on the apex hyperbola. This is useful for some types of whale calls, where the energy is difficult to observe on the T–X plots but very visible on the F–X plot. Click through the F–X plots to ensure none of the whale call energy goes outside of your yellow lines (higher than the top one or lower than the bottom one).

Again, the idea is to capture *all* the whale energy between the four lines that have been placed: the white leading hyperbola, the cyan ending hyperbola, and the minimum and maximum distances marked by the yellow lines. As always, get as close as possible to the whale call without cutting off the energy.



(a) A successful placement of the minimum and maximum cable-distance boundaries. The horizontal yellow lines enclose the whale-call energy.



(b) An incorrectly placed maximum-distance boundary. Some whale-call energy near 63 km lies above the upper yellow line. Drag the line upward, then check the next F–X window as well.

Figure 11: Examples of selecting minimum and maximum cable-distance boundaries.

7 Step 5: Create F–X Bounding Boxes

Press:

`f`

The application enters F–X box-labeling mode. F–X thumbnail windows that correspond to the energy you've identified will be highlighted in the right-side F–X series panel. Pick the one where the whale call is most clearly visible and spans the most frequencies. This will often be the one that is automatically selected when you press `f`, but you may occasionally need to click to an F–X plot slightly later before you get maximum energy. **Make sure you are looking at the correct whale call** – the energy on the F–X plot should align in distance with the energy on the hyperbola for the 2-second window indicated by the gray lines.

Double click somewhere on the F–X plot. This will place a Frequency box. You can resize the box by clicking and dragging the top-left corner of the box, and you can reposition the box by clicking inside the box and dragging it. Drag and resize the box until it aligns with the whale-call

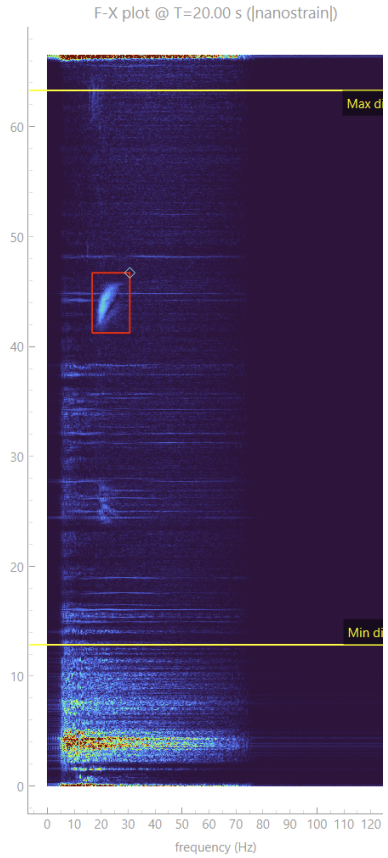


Figure 12: An example of a successfully placed frequency box. The left and right edges closely match the whale call energy.

energy.

Note that the spatial dimension of your frequency box is not important. The GUI will only store the minimum and maximum frequency indicated by the box.

7.0.1 Step 6: Select a Label and Save

The Annotation panel displays all numeric labels, for example:

1	Fin song – 20 Hz
2	Fin feeding – 40 Hz +
3	Blue song – tonal
4	Blue song – pulsed
5	Baleen - unknown
6	Anthropogenic
7	Unknown
8	(not used)
9	Review
0	Cancel

You may either:

- select a label in the dropdown and click **Confirm and Save**; or
- press the corresponding numeric key `1` through `9`.

As soon as you press a number key, the annotation will be saved to the configured CSV file.

(Sidenote: the labels can be customized in the config.py file. We may do this if we observe other calls or annotate other datasets at a future point).

Saved metadata include:

- UTC apex timestamp;
- source-file-relative apex time;
- apex cable distance;
- endpoint-derived duration;
- distance to cable;
- sound speed used for hyperbola geometry;
- minimum/maximum cable distance;
- minimum/maximum F–X frequency;
- label number and label name;
- dataset and source-file metadata;
- save timestamp and username of the annotator.

After saving, temporary annotation overlays are cleared and only the leading curve of the hyperbola (yellow dashed line) and the apex (red dot) are displayed.

7.0.2 Delete, edit, copy + paste

You can **delete** or **edit** previous annotations by clicking on the apex and pressing `delete` or `a` (to re-enter annotation mode) then the key corresponding to the annotation step you would like to edit (`h` for hyperbola, `d` for distance, etc).

You can copy an existing annotation by clicking the apex and pressing `ctrl` + `c`. Then, you can press `ctrl` + `v` and click a new apex point to paste the annotation to a new location. This can be useful when one whale calls repeatedly from the same location several times; you can simply copy/paste rather than redrawing the entire contour!

7.1 Additional functionality

You can select a new spectrogram row by pressing `s` then clicking the row you wish to display. This can be helpful for call identification.

You can toggle the existing annotations on and off using:

Hide Existing Labels
Show Existing Labels

in the Annotation panel.

Existing annotations in the current time window appear in the T–X plot.

7.2 Common Problems

No Endpoint Selected

You pressed `e` to confirm before clicking an endpoint.

`e` → click endpoint → drag if needed → `e`

Two Distance Points Required

You pressed `d` to confirm before selecting both distance boundary points.

`d` → click first point → click second point → `d`

No F–X Boxes Created

You entered F–X mode but did not create a box.

`f` → select relevant F–X thumbnail → double-click main F–X plot → `f`

No F–X Windows Overlap Selected Call Range

Check:

- apex selection;
- endpoint and duration;
- minimum/maximum distance selections;
- F–X window duration.

Endpoint Click Is Too Far From Apex Channel

Click closer to the apex cable-distance row. The endpoint is constrained to the same cable channel as the apex.

8 Quick Keyboard Reference

Key	Function
`a`	Start or confirm apex selection.
`h`	Start or confirm hyperbola distance-to-cable fitting.
`e`	Start or confirm endpoint selection.
`d`	Start or confirm minimum/maximum cable-distance selection.
`f`	Start or finish F–X box annotation.
`Space`	Move to next relevant F–X time window during F–X mode.
`s`	Select cable row for spectrogram.
`1--9`	Assign and save annotation label.
`Esc`	Cancel current annotation.

Key	Function
`click apex`	Delete or edit an existing label.

9 Glossary

Term	Definition
Acoustic wave	A pressure disturbance that travels through a material such as air, water, or the seafloor. Whale calls are acoustic waves travelling through seawater.
Apex	The point on an arrival curve corresponding to the cable location closest to the source. In the annotation interface, the apex is selected first and is used to anchor the hyperbola.
Apparent velocity	The speed at which a signal appears to move along the cable. It may differ from the true sound speed because the source can be some distance away from the cable and because the cable samples only one spatial direction.
Backscatter / Rayleigh backscatter	A small amount of laser light that is naturally scattered backward within an optical fiber because of tiny variations in the glass. DAS instruments analyze changes in this returned light.
Bandpass filter	A processing step that keeps signals within a selected frequency range while reducing signals below and above that range. For example, a 15–30 Hz bandpass retains much of the frequency range relevant to many fin whale 20 Hz calls.
Cable distance	Position measured along the fiber-optic cable, usually in meters or kilometers.
Channel	One spatial measurement location along a DAS cable. A DAS dataset contains many channels, allowing the cable to be treated as a long line of sensors.
Channel spacing	The distance between neighboring DAS channels. Smaller channel spacing provides more closely spaced measurements along the cable, but can increase data size and processing time.
DAS (Distributed Acoustic Sensing)	A measurement technique that uses a fiber-optic cable as a dense array of sensors. A DAS interrogator sends laser pulses through the fiber and measures changes in the returned light caused by stretching or compression of the cable.
Data source	The dataset, interrogator type, selected cable range, channel spacing, and related settings used to load DAS data into the interface.
Distance to cable	The estimated distance between a whale or other acoustic source to the closest portion of the cable. In `annotate_das`, it is adjusted to make a hyperbola align with the observed arrival pattern. It is useful for annotation geometry but is not a precise source-location estimate.
Downsweep	A sound whose frequency decreases over time. Many fin whale 20 Hz calls are downsweeps, often beginning near 25 Hz and ending near 17 Hz.
Endpoint	The selected end time of an annotated call. Together with the apex, the endpoint defines the approximate temporal extent of the event.
F–K filter	A filter operating in frequency–wavenumber space. It can retain or suppress signals based on their apparent velocity along the cable. “F” refers to temporal frequency, and “K” refers to spatial wavenumber.

Term	Definition
F–X plot	A plot of frequency versus cable distance for a short selected time window. In this interface, it helps identify the cable-distance and frequency region containing a call.
F–X thumbnail series	The right-side panel containing a scrollable sequence of F–X plots for short, consecutive time windows across the currently displayed T–X window.
F–X window	A short interval of time used to calculate and display an F–X plot. In `annotate_das`, the default F–X window is typically two seconds long.
Fiber-optic cable	A cable containing thin glass fibers that transmit light. Telecommunications cables can sometimes be used for DAS measurements in addition to their usual communication purpose.
File boundary	The transition between consecutive DAS source-data files. A visible discontinuity or transient can occur at a file boundary when adjacent files have different baseline offsets.
Frequency	The number of repeating cycles of a vibration per second, measured in hertz (Hz). Low-frequency whale calls may occur at tens of hertz; human hearing typically extends to much higher frequencies.
Frequency box	A draggable and resizable box in an F–X plot used to specify the minimum and maximum frequency of a whale call. The interface stores the frequency limits of the box; its spatial width is not used for the saved annotation.
Gauge length	The length of fiber over which a DAS instrument effectively measures changes in strain. It affects the spatial scale over which the DAS measurement is averaged.
GUI (graphical user interface)	The visual application window through which users load data, inspect plots, create annotations, and save labels.
Hertz (Hz)	The unit of frequency. One hertz equals one cycle per second. A 20 Hz fin whale call vibrates approximately 20 times per second.
Hyperbola	A curved shape used in the T–X plot to represent the expected arrival time of sound at different cable positions when a source is located some distance away from the cable.
Interrogator	The DAS instrument that sends laser pulses through the fiber-optic cable and analyzes the returning backscattered light to produce DAS measurements.
Noise	Unwanted signal variation that makes a target sound harder to detect or identify. Noise may come from ocean waves, ships, electronics, cable motion, or data-processing artifacts.
Propagation	The travel of a wave through a medium. For example, a whale call propagates through seawater, reaches the cable at different times, and produces a pattern across DAS channels.
Signal-to-noise ratio (SNR)	A comparison of the strength of a target signal to the background noise. A higher SNR generally makes a call easier to see, detect, and classify.
Sound speed	The speed at which sound travels through a medium. In seawater it is commonly around 1500 meters per second, although it varies with temperature, salinity, and depth/pressure.
Spectrogram	A plot showing how frequency content changes over time. Time is usually on the horizontal axis, frequency is on the vertical axis, and color represents signal strength.
Strain	A fractional change in length: stretching or compression divided by the original length. DAS is primarily sensitive to strain along the direction of the fiber-optic cable.

Term	Definition
Time window	A selected interval of time from a DAS dataset. The T–X plot displays one time window, while an F–X plot is calculated from a shorter time window within it.
T–X plot	A plot of time (T) versus cable distance (X). Sloping or curved features can show how an acoustic arrival reaches different positions along the cable at different times.
UTC	Coordinated Universal Time, the international time standard used for recording and comparing observations from different locations. UTC does not change with local time zones or daylight-saving time.
Whale call	An acoustic signal produced by a whale. In this manual, a whale call is the event being identified, bounded, labeled, and saved as an annotation.
Whale song	An acoustic signal produced by a male whale which serves some sort of function in mating.
Wavelength	The physical distance between equivalent points on successive cycles of a wave, such as peak-to-peak distance. Wavelength depends on both sound speed and frequency.
Wavenumber	A measure of how rapidly a signal changes across space rather than time. It is used in F–K analysis and is related to spatial wavelength.